

Multivariate Time Series Imputation With Transformers

A. Yarkin Yıldız[✉], Emirhan Koç[✉], and Aykut Koç[✉], *Senior Member, IEEE*

Abstract—Processing time series with missing segments is a fundamental challenge that puts obstacles to advanced analysis in various disciplines such as engineering, medicine, and economics. One of the remedies is imputation to fill the missing values based on observed values properly without undermining performance. We propose the Multivariate Time-Series Imputation with Transformers (MTSIT), a novel method that uses transformer architecture in an unsupervised manner for missing value imputation. Unlike the existing transformer architectures, this model only uses the encoder part of the transformer due to computational benefits. Crucially, MTSIT trains the autoencoder by jointly reconstructing and imputing stochastically-masked inputs via an objective designed for multivariate time-series data. The trained autoencoder is then evaluated for imputing both simulated and real missing values. Experiments show that MTSIT outperforms state-of-the-art imputation methods over benchmark datasets.

Index Terms—Deep learning, imputation, multivariate time series, time series, transformer, unsupervised learning.

I. INTRODUCTION

TIME-SERIES data processing is ubiquitous in numerous domains, including healthcare [1], [2], transportation [3], and weather forecasting [4], [5]. In particular, multivariate time-series data demonstrate a dependent sequence over a common independent variable, such as multiple simultaneous sensor recordings used for an autonomous driving maneuver or different channels of a data acquisition device used for clinical diagnosis. In the era of “Big Data,” labeled multivariate time-series datasets are hard to obtain since their acquisition requires extensive time and resources, potentially from trained experts. Moreover, data acquisition failures to collect simultaneous time-series data, such as sensor malfunctions or human errors, have become more prevalent, which causes missing values in data [6], [7]. These missing values hinder data-driven decisions, particularly for deep learning methods that rely on fully-observed

big data. Therefore, there is increasing interest in overcoming the missing patterns to learn from these incomplete data before implementing downstream machine learning models [8], [9].

Two main approaches have typically addressed missing values. One is discarding missing entries or data samples that include missing entries. However, data deletion is not the best approach, as it creates ambiguity in parameter estimation and discards valuable information captured in incomplete samples. The other approach is *imputation*, which estimates missing parts based on the patterns in observed values. Imputation preserves more information and reduces data bias prior to downstream applications. Many prior approaches to imputation involve shallow models, and strong assumptions on missing patterns, including interpolation [10], mean averaging, and k-nearest neighbors [11], [12]. More recently, deep learning-based machine learning approaches have significantly improved performance in deterministic imputation [13], [14], [15] and probabilistic imputation [16], typically via autoregressive modeling assumptions that are prone to compounding errors when generalized to real-life data [17].

Unlike the domination of deep learning in computer vision [18] and language processing [19], a significant body of research has employed shallow machine learning models for time-series analysis [20], [21], [22]. Recently, models using *self-attention* have become a cornerstone in multivariate time-series processing due to their considerable performance superiority over shallow and sophisticated models such as recurrent neural networks (RNN), convolutional neural networks (CNN), and generative-adversarial networks (GAN) [23], [24], [25]. Moreover, transformer architectures have become increasingly popular due to their incorporation of *self-attention* mechanism, which highlights important input features without relying on sequence-aligned convolutions or recurrent models [26]. This allows for successful adaptations to long-term dependencies in time-series data.

Motivated by the recent advances in machine learning for time-series analysis [27], we propose a novel approach called *Multivariate Time-Series Imputation with Transformers* (MTSIT): An unsupervised autoencoder model with a transformer encoder that leverages unlabeled observed data for *joint reconstruction and imputation* of multivariate time-series. We train our autoencoder by stochastically masking and imputing non-missing values using a training objective mainly designed for multivariate time-series data. MTSIT takes a joint reconstruction and imputation approach while employing an autoencoder architecture with a *single transformer encoder* and a linear decoder, significantly reducing computations. Our model has less than 6,000 parameters, unlike cascaded transformer autoencoder architectures with millions of parameters. Unlike recent works, our training objective learns from stochastically-masked data

Manuscript received 21 August 2022; revised 12 November 2022; accepted 18 November 2022. Date of publication 25 November 2022; date of current version 22 December 2022. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Yipeng Liu. (*Corresponding author: Aykut Koç.*)

The authors are with the Department of Electrical and Electronics Engineering, Bilkent University, 06800 Bilkent, Ankara, Turkey, and also with the UMRAM, Bilkent University, 06800 Ankara, Turkey (e-mail: yarkin.yildiz@ug.bilkent.edu.tr; emirhan.koc@bilkent.edu.tr; aykut.koc@bilkent.edu.tr).

This letter has supplementary downloadable material available at <https://doi.org/10.1109/LSP.2022.3224880>, provided by the authors.

Digital Object Identifier 10.1109/LSP.2022.3224880

Source codes are available at <https://github.com/koc-lab/MTSIT>

while performing well on both *simulated and real-life* missing values schemes against benchmark ground-truth data. We experimentally show that MTSIT outperforms seven state-of-the-art imputation methods over benchmark datasets by 13.5–50.5%.

II. TIME-SERIES IMPUTATION VIA NEURAL NETWORKS

Time-series imputation applications have a long-standing history, with deep learning methods gaining popularity in recent years. Che et al. [14] propose a Gated-recurrent Unit (GRU) variant named GRU-D to overcome the missing values in time-series data used for classification. Mulyadi et al. [28] propose a variational recurrent neural network (RNN) that performs imputation using recurrent blocks to exploit temporal relations and deep generative blocks for imputation. Cao et al. [13] create the benchmark framework BRITS, which imputes the missing values by treating them as hidden state variables of a bidirectional RNN. Yoon et al. [29] also impute the missing values via a bidirectional RNN but instead treat missing values as constants. Liu et al. [17] propose an autoencoder model combining a bidirectional encoder and a multiresolution decoder, named NAOMI, for spatio-temporal sequence imputation. Choi et al. [30] propose the Random Drop Imputation with Self-training (RDIS) that learns to impute simulated and randomly missing time-series data. Luo et al. [15] create GRU for imputation (GRUI) that uses a generative adversarial network (GAN). Lastly, Tashiro et al. [31] propose a conditional score-based diffusion model for time-series imputation, named CSDI. Like Gaussian processes, CSDI uses probabilistic imputation and utilizes the correlations between the observed values by explicitly training the conditional diffusion model for imputation and outperforming the state-of-the-art deterministic imputation methods.

Despite the prevalence of the self-attention mechanism in many domains, its application to time-series imputation is still being explored. Suo et al. [32] propose an imputation framework that learns the distant correlations across time and the dependencies of multivariate time series locally and globally using a multi-dimensional attention mechanism. Du et al. [33] propose the method of a self-attention-based mechanism for missing values in multivariate time series (SAITS). SAITS performs joint reconstruction and imputation using a weighted combination of two diagonally-masked self-attention blocks and is extensively evaluated for imputation on simulated missing value patterns on real-life datasets.

III. MTSIT: MULTIVARIATE TIME-SERIES IMPUTATION WITH TRANSFORMERS

We consider N multivariate time-series (MVTs) data samples, each involving T time points and D different variables. Formally, we denote each MVTs sample by $\mathbf{x}_t^{(n)} \in \mathbb{R}^D : \mathbf{X}^{(n)} \in \mathbb{R}^{T \times D} = [\mathbf{x}_1^{(n)}, \mathbf{x}_2^{(n)}, \dots, \mathbf{x}_T^{(n)}]$, for $n \in [1, \dots, N]$. Each MVTs sample $\mathbf{X}^{(n)}$ involves missing values in the set denoted by $M^{(n)} = \{(t, i) \mid \mathbf{x}_{t,i}^{(n)} \text{ is unknown}, t \in \{1, \dots, T\}, i \in \{1, \dots, D\}\}$. More explicitly, M is a list that contains the indices of unknown values. We aim to design an unsupervised method to impute the unknown values in each MVTs sample. To this end, we employ an autoencoder architecture involving a transformer network encoder [26], [27] uniquely designed for MVTs data.

A. Unsupervised Imputation Autoencoder

Each MVTs sample $\mathbf{x}_t$ in $\mathbf{X}$ is first normalized by subtracting the mean μ and dividing by standard deviation σ across the sequence length for each variable. The normalized sequence $\mathbf{X}_{norm}$ is projected into L -dimensional vector space via a linear fully-connected layer and summed with a fully-trainable positional-encoding matrix to preserve the ordering in time series. Projection of $\mathbf{X}_{norm}$ is described as follows:

$$\mathbf{X}_p = \mathbf{X}_{norm} \mathbf{W}_p + \mathbf{B}_p + \mathbf{E}_{pos}, \quad (1)$$

where $\mathbf{W}_p \in \mathbb{R}^{D \times L}$, $\mathbf{B}_p \in \mathbb{R}^{T \times L}$, and $\mathbf{E}_{pos} \in \mathbb{R}^{T \times L}$ are learnable parameters. The projection sequence $\mathbf{X}_p$ constitutes the input sequence of the multi-head self-attention layer (MSA). The MSA block includes three learnable weight matrices that map $\mathbf{X}$ into query $\mathbf{Q}$ of dimension d_k , key $\mathbf{K}$ of dimension d_k and value $\mathbf{V}$ dimension of d_v . The linear mapping is formally described as follows:

$$\mathbf{Q} = \mathbf{X} \mathbf{W}_Q, \mathbf{K} = \mathbf{X} \mathbf{W}_K, \mathbf{V} = \mathbf{X} \mathbf{W}_V, \quad (2)$$

where $\mathbf{W}_Q \in \mathbb{R}^{L \times d_k}$, $\mathbf{W}_K \in \mathbb{R}^{L \times d_k}$ and $\mathbf{W}_V \in \mathbb{R}^{L \times d_v}$. We set $d_k = d_v$ in our work. Attention scores are calculated as scaled-dot product between $\mathbf{Q}$ and $\mathbf{K}$ and the softmax function is applied to find attention weights $\mathbf{A}_{Q,K} \in \mathbb{R}^{T \times T}$. After that, the final output attention $\mathcal{A} \in \mathbb{R}^{T \times d_v}$ map is calculated as a dot product between $\mathbf{A}_{Q,K}$ and $\mathbf{V}$. The resulting output attention maps from each head are concatenated and projected into a sequence $\mathbf{A}_{Q,K,V} \in \mathbb{R}^{T \times L}$ via $\mathbf{W}^O \in \mathbb{R}^{h d_v \times L}$ where h is the number of heads. The entire process is conducted as follows:

$$\mathbf{A}_{Q,K}^i = \text{softmax} \left(\mathbf{Q}^i (\mathbf{K}^i)^T \right), \quad (3)$$

$$\text{head}_i = \mathcal{A}^i = \mathbf{A}_{Q,K}^i \mathbf{V}, \quad (4)$$

$$\mathbf{A}_{Q,K,V} = \mathcal{C}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \mathbf{W}^O, \quad (5)$$

where $\mathcal{C}$ is the concatenation operation and $i \in \{1, 2, \dots, h\}$.

Following the MSA block, $\mathbf{X}_p$ and $\mathbf{A}_{Q,K,V}$ are summed, and batch normalization (BN) is applied. The resulting sequence $\mathbf{X}_{BN} \in \mathbb{R}^{T \times L}$ is fed into a feed-forward network (FFN), and the output sequence $\mathbf{X}_{FFN} \in \mathbb{R}^{T \times L}$ is obtained. Similarly, BN is applied to the summation of $\mathbf{X}_{BN}$ and $\mathbf{X}_{FFN}$, and $\mathbf{X}_{BN}^1 \in \mathbb{R}^{T \times L}$ is obtained as follows:

$$\mathbf{X}_{BN} = \text{BN}(\mathbf{X}_p + \mathbf{A}_{Q,K,V}), \quad (6)$$

$$\text{FFN}(\mathbf{X}_{BN}) = \text{Dropout}(\text{GELU}(\mathbf{X}_{BN} \mathbf{W}_h) \mathbf{W}_o) \quad (7)$$

$$\mathbf{X}_{BN}^1 = \text{BN}(\mathbf{X}_{BN} + \mathbf{X}_{FFN}), \quad (8)$$

where $\mathbf{W}_h \in \mathbb{R}^{L \times H}$ and $\mathbf{W}_o \in \mathbb{R}^{H \times L}$ are weight matrices in FFN layer from input to output through hidden layer, GELU (Gaussian Error Linear Unit) is the nonlinear activation function [34]. Formal definitions of BN and FFN layers are given in [35] and [26], respectively. The transformer consists of N repeating blocks of encoders such that the output sequence of the current block is given as an input sequence to the following encoder block as follows:

$$\mathbf{X}_{BN}^{i+1} = \text{MSA}(\mathbf{X}_{BN}^i), \quad (9)$$

where $i \in \{0, 1, 2, \dots, N-1\}$, $\mathbf{X}_{BN}^0 = \mathbf{X}_p$, and $\text{MSA}(X)$ defines the entire process in a single encoder block.

Unlike the transformer architecture in [26], we use a linear decoder block instead of repetitive decoder blocks with masked

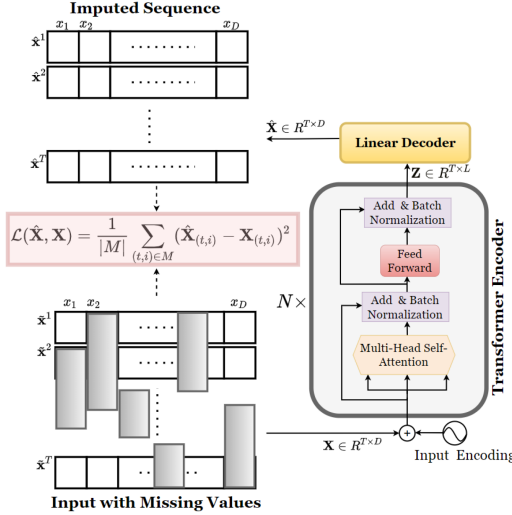


Fig. 1. MTSIT Architecture: Input sequence with missing values (indicated with gray shadows) is given as input to the transformer encoder. The output sequence of the final encoder block is linearly projected into the imputed sequence. MSE is defined between missing value and its imputed prediction.

multi-head self-attention. The output sequence $\mathbf{X}_{BN}^N$ of the last encoder block is fed to the linear decoder and projected into a sequence $\hat{\mathbf{X}} \in \mathbb{R}^{T \times D}$ via learnable projection matrix $\mathbf{W}_o$. In the decoder block, the notation of $\mathbf{X}_{BN}^N$ is replaced with the latent variable $\mathbf{Z}$ and the projection is given:

$$\hat{\mathbf{X}} = \mathbf{Z}\mathbf{W}_o + \mathbf{B}_o. \quad (10)$$

B. Training Objective

To formulate unsupervised learning for imputation, we train our autoencoder by stochastically-masking *non-missing* values and learning to impute them. At evaluation time, the trained autoencoder can impute both real missing and simulated missing values as in training. Fig. 1 illustrates our stochastic masking setup. Each masked time segment length follows a geometric distribution with mean l_m , followed by an unmasked time segment with mean length $l_u = \frac{1-r}{r}l_m$. Each mask is created independently from the other for every training sample and epoch, replicated along variables for consistency, which leads to $r \times m$ masked time points in each step. Unlike masking each time point independently at random, our masking pattern helps the model learn inter-dependencies between the variables and the previous and the following segments in individual variables. We select $l_m = 3$ and $r = 0.15$ for sufficiently long masked segments. Finally, we train our autoencoder to learn to impute the masked time points by minimizing the mean-squared error:

$$\mathcal{L}(\hat{\mathbf{X}}, \mathbf{X}) = \frac{1}{|M|} \sum_{(t,i) \in M} (\hat{\mathbf{X}}_{(t,i)} - \mathbf{X}_{(t,i)})^2, \quad (11)$$

where M is the list that contains the indices of masked values, and $|M|$ denotes the size of the masking list M . In other words, only the predictions corresponding to the masked values are included in the loss calculation for each data sample.

IV. EXPERIMENTS

A. Datasets

We evaluate our approach and the baseline methods using two benchmarks: the Healthcare dataset from the PhysioNet Challenge 2012 [36] and the Beijing Air Quality dataset [37].

1) *Healthcare Physionet Challenge 2012*: The Physionet healthcare dataset consists of clinical data from 4,000 patients for 48 hours with 35 variables, such as blood pressure and heart rate, from the intensive care unit, with different feature variables containing missing data for different patients. Some variables are collected in regular intervals, while some other variables are collected irregularly. Overall, the dataset contains around 80% missing values with no corresponding ground truths. Following the previous works [13], [14], [31], the dataset is processed as hourly time series with 48 consecutive time steps. Data with missing values are split into train and test sets in 80% and 20% proportions, respectively. As the dataset has no ground truth, 10/50/90% of the observed values in the test data are taken as ground truths for which the input data is masked with the Bernoulli distribution. We perform 5-fold cross-validation using this setup and report the average and standard deviation of performances across folds.

2) *Beijing Air Quality*: Beijing air quality dataset consists of hourly measured air quality data of PM25 concentrations from 36 stations in Beijing. Each station holds yearly-observed values, from May 1st, 2014 to April 30th, 2015. There are approximately 13% missing values. Unlike the Healthcare data, this dataset does not have irregularly missing data and, thus, has separate ground-truth values for evaluation. Following the previous studies [13], [31], [32], data are split into two parts: 3rd, 6th, 9th, and 12th months of the year are allocated for the test, and the rest is allocated for the training set. Training is performed via masking and imputing the entries that are not missing in the data allocated for the training set. To prepare ground truths for the test set, for each preceding month of the test data (e.g., data in February for testing March) with the same day and time (e.g., the data of 2015-03-17, 07:00 and 2015-02-17, 07:00), if an entry includes any missing value and the current month (March) does not include any missing value, the regular data of the current month is included in the ground-truth. Then, the corresponding missing location of the former month (February) is indexed as missing inputs and matched with the ground truths in the current month (March). This process is applied for each of the 36 stations, forming hourly time series with 36 consecutive time steps. We repeat experiments with this data 5 times and report the average and standard deviation of performances.

B. Experimental Setup

To design our architecture, we train the autoencoder [27] via Adam optimizer with a learning rate of 0.001 and batch size of 64 for 400 epochs. In our experiments, four attention heads are used in 3 blocks of encoder layers. The hidden unit size of FNN and dimension of latent variable $\mathbf{Z}$ is selected as 16. Learnable positional-encodings are subrogated to the sinusoidal positional-encodings as the experiments show better results in favor of learnable positional-encodings without degrading the computational cost. In this setting, unlike the huge models with millions of parameters, we obtain a scalable model with less than only 6,000 parameters. We trained our models on NVIDIA GeForce GTX 1050 Ti, and the training phases lasted 24 minutes.

TABLE I
RESULTS IN HEALTHCARE AND AIR QUALITY DATASET

Model	Healthcare			Air Quality	
	10% Missing	50% Missing	90% Missing	13%Missing	
	MAE ↓ SE	MAE ↓ SE	MAE ↓ SE	MAE ↓ SE	
V-RIN	0.271 / 0.001	0.365 / 0.002	0.606 / 0.006	25.40 / 0.620	
BRITS	0.284 / 0.001	0.368 / 0.002	0.517 / 0.002	14.11 / 0.260	
RDIS	0.319 / 0.002	0.419 / 0.002	0.631 / 0.002	22.11 / 0.350	
unc-CSDI	0.326 / 0.008	0.417 / 0.010	0.625 / 0.010	11.40 / 0.029	
CSDI	0.217 / 0.001	0.301 / 0.002	0.481 / 0.003	12.13 / 0.07	
GLIMA*	0.265 / -	-	-	10.54 / -	
SAITS*	0.237 / 0.067	0.240 / 0.067	0.262 / 0.069	9.60 / 0.040	
MTSIT*	0.228 / 0.066	0.236 / 0.068	0.238 / 0.068	8.31 / 0.058	

The mean and SE of MAE for five-fold cross-validation and five trials are reported. Self-attention-based methods are indicated with an asterisk (*). The best scores are emboldened.

TABLE II
COMPARISON OF GEOMETRIC VS. BERNOULLI TRAINING OF MTSIT IN HEALTHCARE AND AIR QUALITY DATASETS

Mask	Healthcare			Air Quality	
	10% Missing	50% Missing	90% Missing	13% Missing	
	MAE ↓ SE	MAE ↓ SE	MAE ↓ SE	MAE ↓ SE	
Bernoulli	0.236 / 0.069	0.241 / 0.069	0.262 / 0.073	8.50 / 0.111	
Geometric	0.228 / 0.066	0.236 / 0.068	0.238 / 0.068	8.31 / 0.058	

The mean and SE of MAE for five-fold cross-validation and five trials are reported. The best scores are emboldened.

and 263 minutes at most for the Air Quality and Healthcare datasets, respectively.

MTSIT is compared with seven baseline methods, which are V-RIN [28], BRITS [13], RDIS [30], unconditional CSDI (unc-CSDI), CSDI [31], GLIMA [32], and SAITS [33]. Five-fold cross-validation is reported for the Healthcare dataset, and five trials are reported for the air quality dataset. Each method is evaluated using mean absolute error (MAE) and standard error (SE), where MAE and SE correspond to our results' mean and variance.

C. Results and Discussions

Table I shows the imputation performance of MTSIT against state-of-the-art methods on the Healthcare and Air Quality datasets. Our approach consistently outperforms seven state-of-the-art imputation methods by 13.5–50.5% over all datasets, except the Healthcare, with a missingness rate of 10%, illustrating the capability of MTSIT for learning representative latent features and data distributions. In addition, SAITS and MTSIT exhibit superiority of self-attention in all datasets except the Healthcare, with a 10% missing portion. In Table II, we compare the imputation performance of our autoencoder when trained with the proposed geometric vs. independent Bernoulli distributed masking for each time point. Geometric masking consistently outperforms Bernoulli, further motivating its unique design for MVTS data analysis. We also compare the use of deterministic vs. learnable positional encodings within MTSIT. Results are given in Table III, which shows that learnable positional encodings perform significantly better.

Fig. 2 shows the imputed vs. ground-truth time points on example time windows from the Air Quality and Healthcare

TABLE III
COMPARISON (MAE SCORES) OF DETERMINISTIC VS. LEARNABLE POSITIONAL ENCODINGS OF MTSIT

	Healthcare			Air Quality	
	10% Missing	50% Missing	90% Missing	13% Missing	
	MAE ↓	MAE ↓	MAE ↓	MAE ↓	
Determ.	0.365	0.367	0.371	8.537	
Learn.	0.228	0.236	0.238	8.310	

The best scores are emboldened.

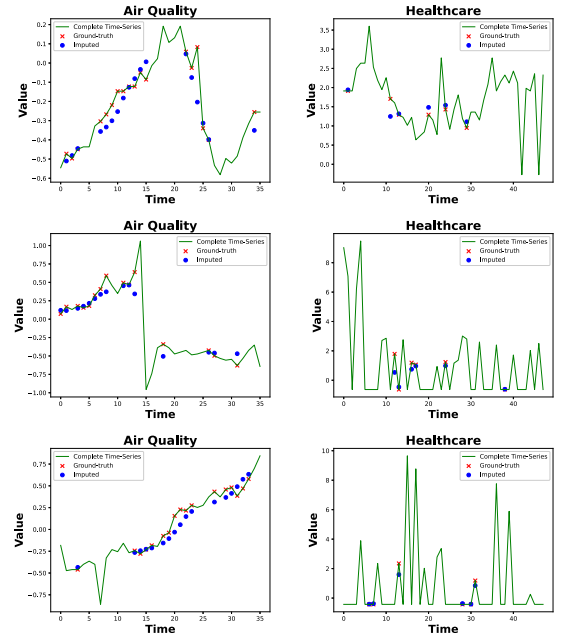


Fig. 2. Examples of the time-series imputation for the Air Quality and Healthcare datasets. The red crosses stand for the masked ground-truth targets, and the blue dots stand for imputed values.

datasets, where ground-truth values correspond to masked missing values. Agreeing with the standard deviations observed in Table I, healthcare imputation occasionally misses sudden spiky trends, which can be suppressed before training by eliminating the powerline noise as a future extension.

V. CONCLUSION

This paper proposes MTSIT, a novel method that uses unsupervised transformer architecture for missing value imputation in multivariate time series. Specifically, an autoencoder imputes both the simulated missing values in training and the real missing values, which is trained by stochastically masking the non-missing values and learning how to impute these values. It is shown that the model outperforms the state-of-the-art imputation methods.

The proposed imputation approach can be extended in future work to incorporate confidence bounds in a Bayesian setting and to design better and advanced MTSIT models by suitably exploiting the recent sophisticated transformer variants such as Autoformer and Informer [38], [39]. Using confidently imputed points to improve performance on forecasting and classification is another promising open direction. Lastly, MTSIT can also be extended to recent efficient transformer variants [40], [41].

REFERENCES

- [1] Z. Hajihashemi and M. Popescu, "A multidimensional time-series similarity measure with applications to eldercare monitoring," *IEEE J. Biomed. Health Inform.*, vol. 20, no. 3, pp. 953–962, May 2016.
- [2] A. Prakosa et al., "Generation of synthetic but visually realistic time series of cardiac images combining a biophysical model and clinical images," *IEEE Trans. Med. Imag.*, vol. 32, no. 1, pp. 99–109, Jan. 2013.
- [3] A. Gupta, H. P. Gupta, B. Biswas, and T. Dutta, "An early classification approach for multivariate time series of on-vehicle sensors in transportation," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 12, pp. 5316–5327, Dec. 2020.
- [4] A. Tascikaraoglu et al., "Compressive spatio-temporal forecasting of meteorological quantities and photovoltaic power," *IEEE Trans. Sustain. Energy*, vol. 7, no. 3, pp. 1295–1305, Jul. 2016.
- [5] Y. Zheng et al., "Forecasting fine-grained air quality based on big data," in *Proc. 21th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2015, pp. 2267–2276.
- [6] X. Jiang, Z. Tian, and K. Li, "A graph-based approach for missing sensor data imputation," *IEEE Sensors J.*, vol. 21, no. 20, pp. 23133–23144, Oct. 2021.
- [7] Y. Zhang, Z. Zhao, Y. Deng, X. Zhang, and Y. Zhang, "Reconstruction of missing samples in antepartum and intrapartum FHR measurements via mini-batch-based minimized sparse dictionary learning," *IEEE J. Biomed. Health Inform.*, vol. 26, no. 1, pp. 276–288, Jan. 2022.
- [8] S. Yang, M. Dong, Y. Wang, and C. Xu, "Adversarial recurrent time series imputation," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Aug. 04, 2020, doi: [10.1109/TNNLS.2020.3010524](https://doi.org/10.1109/TNNLS.2020.3010524).
- [9] S. Z. Dadaneh, E. R. Dougherty, and X. Qian, "Optimal Bayesian classification with missing values," *IEEE Trans. Signal Process.*, vol. 66, no. 16, pp. 4182–4192, Aug. 2018.
- [10] A. T. Sree Dhevi, "Imputing missing values using inverse distance weighted interpolation for time series data," in *Proc. IEEE 6th Int. Conf. Adv. Comput.*, 2014, pp. 255–259.
- [11] Y. Xue, Y. Tang, X. Xu, J. Liang, and F. Neri, "Multi-objective feature selection with missing data in classification," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 6, no. 2, pp. 355–364, Apr. 2022.
- [12] S. Wang and G. Mao, "Missing data estimation for traffic volume by searching an optimum closed cut in urban networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 1, pp. 75–86, Jan. 2019.
- [13] W. Cao, D. Wang, J. Li, H. Zhou, L. Li, and Y. Li, "BRITS: Bidirectional recurrent imputation for time series," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 6774–6784.
- [14] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu, "Recurrent neural networks for multivariate time series with missing values," *Sci. Rep.*, vol. 8, no. 1, pp. 1–12, 2018.
- [15] Y. Luo et al., "Multivariate time series imputation with generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 1603–1614.
- [16] V. Fortuin, D. Baranchuk, G. Rätsch, and S. Mandt, "GP-VAE: Deep probabilistic time series imputation," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2020, pp. 1651–1661.
- [17] Y. Liu, R. Yu, S. Zheng, E. Zhan, and Y. Yue, "NAOMI: Non-autoregressive multiresolution sequence imputation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 11238–11248.
- [18] J. White, T. Kameneva, and C. McCarthy, "Vision processing for assistive vision: A deep reinforcement learning approach," *IEEE Trans. Hum.-Mach. Syst.*, vol. 52, no. 1, pp. 123–133, Feb. 2022.
- [19] D. W. Otter, J. R. Medina, and J. K. Kalita, "A survey of the usages of deep learning for natural language processing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 2, pp. 604–624, Feb. 2021.
- [20] J. Lines, S. Taylor, and A. Bagnall, "Time series classification with HIVE-COTE: The hierarchical vote collective of transformation-based ensembles," *ACM Trans. Knowl. Discov. Data*, vol. 12, no. 5, pp. 1–35, 2018.
- [21] A. Shifaz, C. Pelletier, F. Petitjean, and G. I. Webb, "TS-CHIEF: A scalable and accurate forest algorithm for time series classification," *Data Mining Knowl. Discov.*, vol. 34, no. 3, pp. 742–775, 2020.
- [22] A. Dempster, F. Petitjean, and G. I. Webb, "ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels," *Data Mining Knowl. Discov.*, vol. 34, no. 5, pp. 1454–1495, 2020.
- [23] S. N. Shukla and B. M. Marlin, "Multi-time attention networks for irregularly sampled time series," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [24] E. Oh, T. Kim, Y. Ji, and S. Khyalia, "STING: Self-attention based time-series imputation networks using GAN," in *Proc. IEEE Int. Conf. Data Mining*, 2021, pp. 1264–1269.
- [25] A. Drouin, É. Marcotte and N. Chapados, "TACTiS: Transformer-attentional copulas for time series," in *Proc. 39th Int. Conf. Mach. Learn.*, vol. 162, K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvari, G. Niu and S. Sabato, Eds. Jul. 17–23, 2022, pp. 5447–5493.
- [26] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [27] G. Zerveas, S. Jayaraman, D. Patel, A. Bhamidipaty, and C. Eickhoff, "A transformer-based framework for multivariate time series representation learning," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2021, pp. 2114–2124.
- [28] A. W. Mulyadi, E. Jun, and H.-I. Suk, "Uncertainty-aware variational-recurrent imputation network for clinical time series," *IEEE Trans. Cybern.*, vol. 52, no. 9, pp. 9684–9694, Sep. 2022.
- [29] J. Yoon, W. R. Zame, and M. van der Schaar, "Estimating missing data in temporal data streams using multi-directional recurrent neural networks," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 5, pp. 1477–1490, May 2019.
- [30] T.-M. Choi, J.-S. Kang, and J.-H. Kim, "RDIS: Random drop imputation with self-training for incomplete time series data," in *Proc. 35th AAAI Conf. Artif. Intell.*, 2021.
- [31] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 24804–24816.
- [32] Q. Suo, W. Zhong, G. Xun, J. Sun, C. Chen, and A. Zhang, "GLIMA: Global and local time series imputation with multi-directional attention learning," in *Proc. IEEE Int. Conf. Big Data*, 2020, pp. 798–807.
- [33] W. Du, D. Côté, and Y. Liu, "SAITS: Self-attention-based imputation for time series," 2022, *arXiv:2202.08516*.
- [34] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.
- [35] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
- [36] I. Silva, G. Moody, D. J. Scott, L. A. Celi, and R. G. Mark, "Predicting in-hospital mortality of ICU patients: The physionet/computing in cardiology challenge 2012," in *Proc. IEEE Comput. Cardiol.*, 2012, pp. 245–248.
- [37] S. Zhang, B. Guo, A. Dong, J. He, Z. Xu, and S. X. Chen, "Cautionary tales on air-quality improvement in Beijing," *Proc. Roy. Soc. A: Math., Phys. Eng. Sci.*, vol. 473, no. 2205, 2017, Art. no. 20170457.
- [38] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 22419–22 430.
- [39] H. Zhou et al., "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 12, pp. 11106–11115.
- [40] J. Lee-Thorp, J. Ainslie, I. Eckstein, and S. Ontanon, "FNet: Mixing tokens with Fourier transforms," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Hum. Lang. Technol.*, 2022, pp. 4296–4313.
- [41] F. Sahinuc and A. Koç, "Fractional Fourier transform meets transformer encoder," *IEEE Signal Process. Lett.*, vol. 29, pp. 2258–2262, 2022.